

# Glengarry Pl upper deck

## · 1 · Fulton rail · existing columns and caissons

166 Glengarry Pl, Castle Rock · Vintage Coastline, Dark Hickory picture frame and breakers · SYP @ 12" · 6x12 DF beams · Cortex

| CHECK OR ACH     | FINANCED         | MONTHLY        | SQUARE FEET  | FOOTINGS | RAIL           |
|------------------|------------------|----------------|--------------|----------|----------------|
| <b>\$128,893</b> | <b>\$138,595</b> | <b>\$1,608</b> | <b>741.5</b> | <b>9</b> | <b>69.7 LF</b> |

### DESIGN READS

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frame</b>  | 7 zones, 51'-4" along the house, 26'-6" at the deepest                                                                                        |
| <b>Deck</b>   | 74.0 LF of outer edge · 741.5 SF · 12'-0" above grade                                                                                         |
| <b>Zone</b>   | L Stair pocket — 4'-4" x 4'-4" past the house's west wall; the stair leaves its north edge along the wall: 4'-4" x 4'-4", wall set back 4'-4" |
| <b>Zone</b>   | P West pocket — 2'-4" beside the bay, ledgered to the recessed wall: 2'-4" x 10'-4", wall set back 4'-4"                                      |
| <b>Zone</b>   | P2 West strip — 4'-3" from the recessed wall to the curved front: 4'-3" x 16'-10", wall set back 4'-4"                                        |
| <b>Zone</b>   | W In front of the bay window — 7'-4" x 5', the bay projects 6' (multi-angle facets in the model): 7'-4" x 6'-6"                               |
| <b>Zone</b>   | M Main run — 19'-11" wall, 11' deep: 19'-11" x 12'-6"                                                                                         |
| <b>Zone</b>   | S Step — 2'-1" at the 7'-6" jog: 2'-1" x 18'-6", wall set back 7'-6"                                                                          |
| <b>Zone</b>   | A Alcove bay — 11'-1" into the 17' alcove, 26'-6" deep, ends at the east wing wall: 11'-1" x 26'-6", wall set back 15'-6"                     |
| <b>Joists</b> | 57 2x12 #1 True Frame Joist SYP GC @ 12" OC                                                                                                   |
| <b>Rims</b>   | 2-ply 2x12 front and sides and rear (freestanding)                                                                                            |
| <b>Beam</b>   | DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0 1/2" / 5'-7 1/2" (worst 1,394 lb)                                                                    |
| <b>Beam</b>   | DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6" (worst 5,409 lb)                                        |
| <b>Beam</b>   | DROP 6x12 DF A: 11'-0", 2 posts at 41'-3 1/2" / 50'-3 1/2" (worst 3,320 lb)                                                                   |

|         |                                                                                                                                                         |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Posts   | the 9 existing stucco columns and caissons stay; new post bases epoxy-set on the caissons                                                               |
| Load    | 50 psf (40 psf live governs) · engineered                                                                                                               |
| Decking | TimberTech Vintage Coastline — 55 rows parallel to the house, picture frame & dividers in Vintage Dark Hickory, 3/16" gaps, Cortex                      |
| Rail    | Fulton 36" Black: 13 bays, 17 posts, 69.7 LF                                                                                                            |
| Stair   | wall:L: 13 risers @ 7.38", 11 treads, 5 stringers, 4'-0" wide — 7 risers + 6 risers (left); landings 4'-0" x 4'-0" + 7'-8" x 5'-8"; landing guard 21 LF |

- Rows per panel: panel 1 (0"-18'-3") 34 rows · panel 2 (18'-3"-38'-2") 26 rows · panel 3 (38'-2"-51'-4") 55 rows · 3 ripped pieces · 2 breakers
- 3 slivers under 6" beside the breakers / bay are not laid as boards (the border or breaker covers)
- DROP 6x12 DF L+P: 6'-7" long, 2 posts at 1'-0 1/2" / 5'-7 1/2"
- DROP 6x12 DF P2+W+M+S+A: 44'-7" long, 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6"; pieces 11'-9 1/2" · 20'-0" · 12'-9 1/2"
- DROP 6x12 DF A: 11'-0" long, 2 posts at 41'-3 1/2" / 50'-3 1/2"
- L: freestanding zone framed on its declared beam; no ledger — lateral bracing per the engineer
- P: 6'-8 3/4" cantilever past the drop beam exceeds L/4 of the 3'-5 3/4" back-span — engineered condition (owner-directed)
- P2: 3'-8 3/4" cantilever past the drop beam exceeds L/4 of the 12'-11 3/4" back-span — engineered condition (owner-directed)
- W: 3'-8 3/4" cantilever past the drop beam exceeds L/4 of the 2'-7 3/4" back-span — engineered condition (owner-directed)
- M: 3'-8 3/4" cantilever past the drop beam exceeds L/4 of the 8'-7 3/4" back-span — engineered condition (owner-directed)
- A: added a drop beam at 12'-2 3/8" from the house — 2x12 @ 12" OC SYP only spans 18'-0"

SITE & CODE FLAGS internal

|          |             |                                                                                                                                                                   |
|----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENGINEER | ENGINEERING | DROP 6x12 DF: 6x12 not in IRC R507.5 — treated as (3)2x12; engineer to confirm; 6x12 not in IRC R507.5 — treated as (3)2x12; engineer to confirm IRC Table R507.5 |
| ENGINEER | ENGINEERING | Cantilever 8 3/4" > L/4 of the 0" back-span — owner-directed, in the stamped set IRC R507.5 / R507.6                                                              |
| ENGINEER | ENGINEERING | Deck surface 12'-0" above grade — over 10': lateral bracing and an engineered design                                                                              |
| ENGINEER | FOOTING     | existing caissons reused — the engineer verifies capacity for the new loads (5,409 lb worst post); core or probe if no record IRC R507.3                          |
| ENGINEER | ENGINEERING | existing stucco columns reused — the engineer verifies the column core, its connection to the caisson, and details the new column caps                            |
| CODE     | ENGINEERING | posts over 8' (9'-5 1/2") — add diagonal knee bracing at each post (both directions at the corners) DCA6 / IRC R507.4                                             |

|       |             |                                                                                                                                                                                                                               |
|-------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CODE  | ENGINEERING | freestanding deck — no ledger lateral tie; provide diagonal bracing at posts or an engineered lateral system IRC R507.9 / R507.4                                                                                              |
| CODE  | FIRE        | WUI zone: Vintage is Ignition Resistant, Class A, WUI — keep the under-deck area clear of combustibles, 1-hr or noncombustible enclosure may be required IWUIC 504.7                                                          |
| CODE  | STAIRS      | wall:L stair: 13 risers — graspable handrail 34–38" on at least one side (IRC R311.7.8) IRC R311.7                                                                                                                            |
| CODE  | STAIRS      | wall:L stair landing: 36" min in the direction of travel and at least as wide as the stair; guard on every open side over 30" (21 LF counted) IRC R311.7.6 / R312.1                                                           |
| CHECK | SETBACK     | rear setback not entered — confirm the required rear setback and the deck-to-line distance before the frame is laid out zoning                                                                                                |
| CHECK | SETBACK     | side setback not entered — confirm the required side setback and the deck-to-line distance before the frame is laid out zoning                                                                                                |
| INFO  | WIND        | basic wind speed 115 mph — standard connectors                                                                                                                                                                                |
| INFO  | FIRE        | Fulton rail is noncombustible (Colorado Wildfire Resiliency Code practice: Class A decking, noncombustible rail, metal flashing at every wall)                                                                                |
| INFO  | STAIRS      | wall:L stair: flights: 7 risers / 6 treads, run 5.8' · 6 risers / 5 treads, run 4.9' — left at the landing; landing guard 21 LF IRC R311.7                                                                                    |
| INFO  | STAIRS      | wall:L stair: 13 risers @ 7.38" · 11 treads @ 11.66" · run 10'-8 1/4" · stringers 5 @ 12" OC (composite treads) — 7 risers (run 5'-9 15/16") then 6 risers (run 4'-10 5/16") with a landing between IRC R311.7.5 / TimberTech |

THE ORDER   Decks & Docks · GSEXT1 · ship to the job

Lumber

| NET | ORDER | UNIT | ITEM                               | WHY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12  | 13    | ea   | 2x12x10 #1 True Frame Joist SYP GC | +1 cull (field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · 5 stair stringer)                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 30  | 31    | ea   | 2x12x12 #1 True Frame Joist SYP GC | +1 cull (4 front rim ply · ledger · ledger, rear rim ply · rear rim ply, stair stringer · 2 front rim ply, stair stringer · ledger, stair stringer · blocking, field joist, stair stringer · 2 blocking, field joist · 2 PF joist, blocking, field joist · 2 blocking, field joist, left rim ply · 2 PF joist, blocking, right rim ply · blocking, front rim ply, left rim ply · blocking, front rim ply, ledger, left rim ply · blocking, front rim ply, ledger, right rim ply · blocking, right rim ply · 7 blocking) |
| 26  | 27    | ea   | 2x12x14 #1 True Frame Joist SYP GC | +1 cull (19 field joist · 2 PF joist · divider doubler · 2 left rim ply · 2 right rim ply)                                                                                                                                                                                                                                                                                                                                                                                                                              |

| NET | ORDER     | UNIT | ITEM                               | WHY                                                                                                                  |
|-----|-----------|------|------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 20  | <b>21</b> | ea   | 2x12x20 #1 True Frame Joist SYP GC | +1 cull (4 field joist · 4 PF joist · 4 left rim ply · 4 right rim ply · 2 front rim ply · ledger · divider doubler) |
| 16  | <b>17</b> | ea   | 2x12x24 #1 True Frame Joist SYP GC | +1 cull (10 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply)                                             |
| 2   | <b>3</b>  | ea   | 2x6x12 #1 True Frame Joist SYP GC  | +1 cull (2 stair kicker/hanger board)                                                                                |
| 3   | <b>3</b>  | ea   | 6x12x12 Douglas Fir #2             | exact (3 drop beam: DROP 6x12 DF L+P · DROP 6x12 DF P2+W+M+S+A · DROP 6x12 DF A)                                     |
| 1   | <b>1</b>  | ea   | 6x12x20 Douglas Fir #2             | exact (drop beam: DROP 6x12 DF L+P · DROP 6x12 DF P2+W+M+S+A · DROP 6x12 DF A)                                       |
| 1   | <b>1</b>  | ea   | 6x12x16 Douglas Fir #2             | exact (drop beam: DROP 6x12 DF L+P · DROP 6x12 DF P2+W+M+S+A · DROP 6x12 DF A)                                       |

### Hardware

| NET | ORDER      | UNIT | ITEM                                                           | WHY                                                               |
|-----|------------|------|----------------------------------------------------------------|-------------------------------------------------------------------|
| 114 | <b>114</b> | ea   | Simpson LUS210Z face-mount hanger 2x10/2x12                    | exact (57 joists x 2 ends)                                        |
| 28  | <b>28</b>  | ea   | Simpson HUCQ212-2-SDS concealed-flange double 2x12 hanger      | exact (2 side rims x 2 ends) — confirm stock                      |
| 3   | <b>3</b>   | box  | Camo 9ga HDG hanger nail 3" 5 lb (~350)                        | 912 needed of ~1050                                               |
| 3   | <b>3</b>   | box  | Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)                    | 1534 needed of ~1950                                              |
| 1   | <b>1</b>   | box  | FastenMaster LedgerLOK 3-5/8" 50 ct                            | 48 needed                                                         |
| 28  | <b>28</b>  | ea   | Simpson DTT1Z lateral tie kit w/ SDS screws                    | exact (2 near each end of every ledger, into house floor framing) |
| 85  | <b>85</b>  | ea   | Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden) | exact (57 joists + 2 rims at each drop beam)                      |
| 9   | <b>9</b>   | ea   | Simpson CCQ68SDS2.5 column cap 6x beam / 8x post               | exact                                                             |
| 1   | <b>1</b>   | box  | Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct        | 90 needed (bases 0 · caps 90)                                     |
| 10  | <b>10</b>  | box  | GRK RSS #10 x 3-1/8" structural screw 100 ct                   | 998 needed (laminations 554 · blocking 444)                       |
| 46  | <b>48</b>  | ea   | 7/16" x 4-1/2" HDG hex bolt + nut + 2 washers                  | +2 (23 rail posts x 2)                                            |
| 10  | <b>10</b>  | ea   | Simpson LSCZ adjustable stair-stringer connector               | exact (wall:L stair, 1 per stringer at the rim — 2 flights)       |

### Flashing & waterproofing

| NET | ORDER    | UNIT | ITEM                                                                           | WHY                                                       |
|-----|----------|------|--------------------------------------------------------------------------------|-----------------------------------------------------------|
| 2   | <b>2</b> | roll | Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims) | 79 LF of house wall with laps — nothing on a privacy wall |
| 8   | <b>8</b> | ea   | Aluminum L-flashing 10' (over ledger + side rims)                              | house walls only                                          |

| NET | ORDER     | UNIT | ITEM                                                                                                                            | WHY                             |
|-----|-----------|------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 8   | <b>8</b>  | ea   | 26 ga galvanized wall flashing 12" x 10' (CWRC 404.3.2: noncombustible flashing at the deck/wall intersection, behind cladding) | WUI practice — house walls only |
| 16  | <b>16</b> | roll | Nichigo G-Tape 3035 2" x 65' black                                                                                              | 977' needed of 1040'            |
| 5   | <b>5</b>  | roll | Nichigo G-Tape 3035 4" x 65' black                                                                                              | 303' needed of 325'             |

### Decking

| NET | ORDER     | UNIT | ITEM                                                                    | WHY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|-----------|------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30  | <b>32</b> | ea   | TimberTech Vintage Coastline 1x6x12 Square Edge                         | +2 (30 x 12' · 30 x 16' · 40 x 20' (17 cut into 2-3 rows) · panel 1 34 rows · panel 2 26 rows · panel 3 55 rows · every row one piece)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30  | <b>32</b> | ea   | TimberTech Vintage Coastline 1x6x16 Square Edge                         | +2 (30 x 12' · 30 x 16' · 40 x 20' (17 cut into 2-3 rows) · panel 1 34 rows · panel 2 26 rows · panel 3 55 rows · every row one piece)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 40  | <b>42</b> | ea   | TimberTech Vintage Coastline 1x6x20 Square Edge                         | +2 (30 x 12' · 30 x 16' · 40 x 20' (17 cut into 2-3 rows) · panel 1 34 rows · panel 2 26 rows · panel 3 55 rows · every row one piece)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1   | <b>1</b>  | ea   | TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders & dividers | exact (border 24'-8",12'-4 9/16" 2'-8 1/2" + border 4'-4",6'-0" 2'-4 15/16" + border 20'-2",12'-5 5/8" 2'-4" + border 22'-5",12'-5 5/8" 2'-4")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5   | <b>6</b>  | ea   | TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders & dividers | +1 (breaker DM/S 19'-11 5/8" · border 38'-2",11'-0" 13'-2 15/16" + border 4'-4",0" 6'-1" · breaker DW/M 12'-7" + border 6'-8",6'-0" 5'-1" · border 0",-4'-4" 4'-4 15/16" + border 0",0" 4'-4 15/16" + border 35'-11",11'-0" 2'-8 15/16" + border 6'-8",11'-0" 2'-8 15/16" + border 8'-11",11'-4 13/16" 2'-8 13/16" + border 33'-8",11'-4 13/16" 2'-8 13/16" · border 11'-2",11'-8 7/8" 2'-8 11/16" + border 31'-5",11'-8 7/8" 2'-8 11/16" + border 13'-5",12'-0 1/8" 2'-8 5/8" + border 29'-2",12'-0 1/8" 2'-8 5/8" + border 15'-8",12'-2 3/4" 2'-8 9/16" + border 26'-11",12'-2 3/4" 2'-8 9/16" + border 17'-11",12'-4 9/16" 2'-8 1/2") |

### Fasteners

| NET | ORDER     | UNIT | ITEM                                                               | WHY                                                              |
|-----|-----------|------|--------------------------------------------------------------------|------------------------------------------------------------------|
| 10  | <b>10</b> | box  | Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)       | 3454 screws (2 per board at every bearing point) · 3500 supplied |
| 1   | <b>1</b>  | box  | Cortex for TimberTech PVC, Dark Hickory plugs — borders + dividers | 232 screws                                                       |

### Fascia

| NET | ORDER | UNIT | ITEM                                             | WHY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------|------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21  | 22    | ea   | TimberTech Vintage Fascia Coastline 0.5x11.75x12 | +1 (left fascia (L) 4'-4 1/2" · right fascia (L) 4'-4 1/2" · front fascia (L) 4'-5" · left fascia (P) 10'-4 1/2" · right fascia (P) 10'-4 1/2" · front fascia (P) 2'-5" · left fascia (P2) 12'-0" · left fascia (P2) 4'-10 1/2" · right fascia (P2) 12'-0" · right fascia (P2) 4'-10 1/2" · front fascia (P2) 4'-4" · left fascia (W) 6'-6 1/2" · right fascia (W) 6'-6 1/2" · front fascia (W) 7'-5" · left fascia (M) 12'-0" · left fascia (M) 6 1/2" · right fascia (M) 12'-0" · right fascia (M) 6 1/2" · front fascia (M) 12'-0" · front fascia (M) 8'-0" · left fascia (S) 12'-0" · left fascia (S) 6'-6 1/2" · right fascia (S) 12'-0" · right fascia (S) 6'-6 1/2" · front fascia (S) 2'-2" · left fascia (A) 12'-0" · left fascia (A) 12'-0" · left fascia (A) 2'-6 1/2" · right fascia (A) 12'-0" · right fascia (A) 12'-0" · right fascia (A) 2'-6 1/2" · front fascia (A) 11'-2") |

### Rail

| NET | ORDER | UNIT | ITEM                                                                                          | WHY                                                                    |
|-----|-------|------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 7   | 7     | ea   | TimberTech Fulton Rail 6' x 36" level panel Black                                             | exact (cut to 1'-8 1/2" / 10 1/2" / 3'-8 1/2" / 5'-4 1/2" / 5'-5 3/4") |
| 6   | 6     | ea   | TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay) | exact (cut to 5'-10 1/2" / 6'-2" / 6'-4 5/16")                         |
| 4   | 4     | ea   | TimberTech Fulton 2" CORNER post 36" Black w/ brackets, cap, skirt                            | exact                                                                  |
| 8   | 8     | ea   | TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt                               | exact                                                                  |
| 5   | 5     | ea   | TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt                              | exact                                                                  |
| 1   | 1     | ea   | Rust-Oleum flat black touch-up (cut ends)                                                     | exact                                                                  |

### Stairs

| NET | ORDER | UNIT | ITEM                                                                               | WHY                                            |
|-----|-------|------|------------------------------------------------------------------------------------|------------------------------------------------|
| 5   | 5     | ea   | 2x12x12 #1 True Frame Joist SYP GC — landing 1 frame (4'-0" x 4'-0", wall:L stair) | +1 (rims + 6 joists @ 12")                     |
| 4   | 4     | ea   | 8x8x8 — landing 1 posts (4)                                                        | exact                                          |
| 4   | 4     | ea   | Landing 1 footings — 4 concrete piers 12" x frost, wet-set bases                   | exact                                          |
| 4   | 4     | ea   | TimberTech Vintage Coastline 1x6x12 — landing 1 decking (16 SF)                    | +10% cuts                                      |
| 8   | 8     | ea   | 2x12x12 #1 True Frame Joist SYP GC — landing 2 frame (7'-8" x 5'-8", wall:L stair) | +1 (rims + 9 joists @ 12")                     |
| 4   | 4     | ea   | 8x8x8 — landing 2 posts (4)                                                        | exact                                          |
| 4   | 4     | ea   | Landing 2 footings — 4 concrete piers 12" x frost, wet-set bases                   | exact                                          |
| 9   | 9     | ea   | TimberTech Vintage Coastline 1x6x12 — landing 2 decking (43 SF)                    | +10% cuts                                      |
| 11  | 12    | ea   | TimberTech Vintage Coastline 1x6x12 Square Edge — treads (wall:L stair)            | +1 (11 treads x 2 boards @ 4'-0", 2 per board) |
| 7   | 7     | ea   | TimberTech Vintage Coastline riser board 7-1/4" x 12' (wall:L stair)               | exact (13 risers @ 4'-0")                      |
| 2   | 2     | ea   | TimberTech Fulton Rail 6' x 36" STAIR panel Black (wall:L stair)                   | exact                                          |

| NET | ORDER | UNIT | ITEM                                                                                    | WHY                                               |
|-----|-------|------|-----------------------------------------------------------------------------------------|---------------------------------------------------|
| 1   | 1     | ea   | TimberTech Fulton Rail 8' x 36" STAIR panel Black (wall:L stair)                        | exact                                             |
| 6   | 6     | ea   | TimberTech Fulton 2" STAIR post Black w/ brackets, cap (wall:L stair)                   | exact (top + bottom of each rail side, 2 flights) |
| 3   | 3     | ea   | TimberTech Fulton Rail 8' x 36" LEVEL panel Black — landing guard (wall:L stair, 21 LF) | exact (open sides of the landings)                |
| 5   | 5     | ea   | TimberTech Fulton 2" post Black w/ brackets, cap — landing guard (wall:L stair)         | exact (bay ends + turn corners)                   |
| 1   | 1     | ea   | Fulton graspable handrail kit (wall:L stair)                                            | exact (4+ risers — IRC R311.7.8)                  |

### Site

| NET | ORDER | UNIT | ITEM                          | WHY                                         |
|-----|-------|------|-------------------------------|---------------------------------------------|
| 700 | 700   | SF   | Demo existing deck + haul-off | labor line — field verify the existing deck |

### FASTENER & CONNECTOR SCHEDULE

|            |                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ledger     | 2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners                                    |
| Joist tape | G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)                                                               |
| Field      | Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (3454 field + 232 border/divider) |
| Rail       | Fulton 36" — 17 posts (4 CORNER, 8 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post                      |

### ▼ Cut list

| MEMBER                            | STOCK | LENGTH     | QTY | NOTE                                                           |
|-----------------------------------|-------|------------|-----|----------------------------------------------------------------|
| Field joists (L)                  | 2x12  | 3'-10"     | 3   | @ 12" OC, crown up                                             |
| Picture-frame joists (L)          | 2x12  | 3'-10"     | 2   | centre 5 1/4" from each frame face                             |
| Side rim plies (L)                | 2x12  | 3'-10"     | 4   | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered |
| Front rim plies (L)               | 2x12  | 4'-4"      | 2   | inner ply takes the hangers; outer ply laminated after         |
| Rear rim plies (freestanding) (L) | 2x12  | 4'-4"      | 2   |                                                                |
| Blocking (L)                      | 2x12  | 10 1/2"    | 8   | 2 rows at 3'-7 1/4", 3'-7 1/4" from the wall                   |
| Field joists (P)                  | 2x12  | 9'-11 1/2" | 1   | @ 12" OC, crown up                                             |
| Picture-frame joists (P)          | 2x12  | 9'-11 1/2" | 2   | centre 5 1/4" from each frame face                             |
| Side rim plies (P)                | 2x12  | 9'-11 1/2" | 4   | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered |

| MEMBER                    | STOCK | LENGTH     | QTY | NOTE                                                                    |
|---------------------------|-------|------------|-----|-------------------------------------------------------------------------|
| Front rim plies (P)       | 2x12  | 2'-4"      | 2   | inner ply takes the hangers; outer ply laminated after                  |
| Ledger (P)                | 2x12  | 2'-4"      | 1   | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (P)              | 2x12  | 10 1/2"    | 4   | 2 rows at 3'-7 1/4", 1'-10 3/8" from the wall                           |
| Field joists (P2)         | 2x12  | 16'-5 1/2" | 3   | @ 12" OC, crown up                                                      |
| Picture-frame joists (P2) | 2x12  | 16'-5 1/2" | 2   | centre 5 1/4" from each frame face                                      |
| Side rim plies (P2)       | 2x12  | 16'-5 1/2" | 4   | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered          |
| Front rim plies (P2)      | 2x12  | 4'-3"      | 2   | inner ply takes the hangers; outer ply laminated after                  |
| Ledger (P2)               | 2x12  | 4'-3"      | 1   | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (P2)             | 2x12  | 10 1/2"    | 8   | 2 rows at 13'-1 1/4", 6'-7 3/8" from the wall                           |
| Field joists (W)          | 2x12  | 6'-1 1/2"  | 6   | @ 12" OC, crown up                                                      |
| Picture-frame joists (W)  | 2x12  | 6'-1 1/2"  | 2   | centre 5 1/4" from each frame face                                      |
| Side rim plies (W)        | 2x12  | 6'-1 1/2"  | 4   | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered          |
| Front rim plies (W)       | 2x12  | 7'-4"      | 2   | inner ply takes the hangers; outer ply laminated after                  |
| Ledger (W)                | 2x12  | 7'-4"      | 1   | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (W)              | 2x12  | 10 1/2"    | 14  | 2 rows at 2'-9 1/4", 1'-5 3/8" from the wall                            |
| Field joists (M)          | 2x12  | 12'-1 1/2" | 19  | @ 12" OC, crown up                                                      |
| Picture-frame joists (M)  | 2x12  | 12'-1 1/2" | 2   | centre 5 1/4" from each frame face                                      |
| Divider doublers (M)      | 2x12  | 12'-1 1/2" | 1   | sister a joist on the divider line so both board edges bear             |
| Side rim plies (M)        | 2x12  | 12'-1 1/2" | 4   | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered          |
| Front rim plies (M)       | 2x12  | 19'-11"    | 2   | inner ply takes the hangers; outer ply laminated after                  |
| Ledger (M)                | 2x12  | 19'-11"    | 1   | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (M)              | 2x12  | 10 1/2"    | 40  | 2 rows at 8'-9 1/4", 4'-5 3/8" from the wall                            |
| Field joists (S)          | 2x12  | 18'-1 1/2" | 1   | @ 12" OC, crown up                                                      |
| Picture-frame joists (S)  | 2x12  | 18'-1 1/2" | 2   | centre 5 1/4" from each frame face                                      |
| Divider doublers (S)      | 2x12  | 18'-1 1/2" | 1   | sister a joist on the divider line so both board edges bear             |
| Side rim plies (S)        | 2x12  | 18'-1 1/2" | 4   | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered          |
| Front rim plies (S)       | 2x12  | 2'-1"      | 2   | inner ply takes the hangers; outer ply laminated after                  |
| Ledger (S)                | 2x12  | 2'-1"      | 1   | top at deck height less board thickness; membrane behind, flashing over |

| MEMBER                     | STOCK             | LENGTH     | QTY | NOTE                                                                      |
|----------------------------|-------------------|------------|-----|---------------------------------------------------------------------------|
| Blocking (S)               | 2x12              | 10 1/2"    | 4   | 2 rows at 16'-3 1/4", 8'-2 3/8" from the wall                             |
| Field joists (A)           | 2x12              | 26'-1 1/2" | 10  | @ 12" OC, crown up                                                        |
| Picture-frame joists (A)   | 2x12              | 26'-1 1/2" | 2   | centre 5 1/4" from each frame face                                        |
| Side rim plies (A)         | 2x12              | 26'-1 1/2" | 4   | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered            |
| Front rim plies (A)        | 2x12              | 11'-1"     | 2   | inner ply takes the hangers; outer ply laminated after                    |
| Ledger (A)                 | 2x12              | 11'-1"     | 1   | top at deck height less board thickness; membrane behind, flashing over   |
| Blocking (A)               | 2x12              | 10 1/2"    | 33  | 3 rows at 12'-2 3/8", 24'-3 1/4", 6'-1 15/16" from the wall               |
| Field boards panel 1       | Vintage Coastline | 7"         | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 9"         | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 11"        | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 1'-0"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 1'-7"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 2'-3"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 2'-11"     | 1   | one board wall to border, no joints                                       |
| Field boards panel 1 (rip) | Vintage Coastline | 3'-0"      | 1   | one board wall to border, no joints; ripped to fit the border / the house |
| Field boards panel 1       | Vintage Coastline | 6'-2"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 6'-4"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 6'-5"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 6'-7"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 6'-9"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 6'-11"     | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 7'-0"      | 2   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 7'-2"      | 2   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 7'-4"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 7'-5"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 8'-0"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 8'-8"      | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 10'-1"     | 1   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 10'-6"     | 8   | one board wall to border, no joints                                       |
| Field boards panel 1       | Vintage Coastline | 10'-11"    | 10  | one board wall to border, no joints                                       |
| Field boards panel 2 (rip) | Vintage Coastline | 10'-0"     | 1   | one board wall to border, no joints; ripped to fit the border / the house |

| MEMBER                             | STOCK                       | LENGTH      | QTY | NOTE                                                                                                                          |
|------------------------------------|-----------------------------|-------------|-----|-------------------------------------------------------------------------------------------------------------------------------|
| Field boards panel 2               | Vintage Coastline           | 14'-8"      | 1   | one board wall to border, no joints                                                                                           |
| Field boards panel 2               | Vintage Coastline           | 17'-10"     | 1   | one board wall to border, no joints                                                                                           |
| Field boards panel 2               | Vintage Coastline           | 19'-5"      | 22  | one board wall to border, no joints                                                                                           |
| Field boards panel 2 (rip)         | Vintage Coastline           | 19'-5"      | 1   | one board wall to border, no joints; ripped to fit the border / the house                                                     |
| Field boards panel 3               | Vintage Coastline           | 11'-1"      | 17  | one board wall to border, no joints                                                                                           |
| Field boards panel 3               | Vintage Coastline           | 12'-11"     | 38  | one board wall to border, no joints                                                                                           |
| Slivers under 6" (not laid)        | Vintage Coastline           | 0"          | 3   | beside the breakers at the bay — the breaker / border covers                                                                  |
| Stair stringers (wall:L, flight 1) | 2x12                        | 8'-3"       | 5   | 7 risers @ 7.385" · 6 treads @ 11.66" · stringers 12" OC                                                                      |
| Stair stringers (wall:L, flight 2) | 2x12                        | 7'-1 3/16"  | 5   | 6 risers @ 7.385" · 5 treads @ 11.66" · stringers 12" OC                                                                      |
| DROP 6x12 DF L+P beam              | 6x12                        | 6'-7"       | 1   | y -8 13/16"; posts at 1'-0 1/2" / 5'-7 1/2"                                                                                   |
| DROP 6x12 DF P2+W+M+S+A beam       | 6x12                        | 44'-7"      | 3   | y 8'-9 3/16"; posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6"; pieces 11'-9 1/2" · 20'-0" · 12'-9 1/2", spliced over posts |
| DROP 6x12 DF A beam                | 6x12                        | 11'-0"      | 1   | y -3'-3 5/8"; posts at 41'-3 1/2" / 50'-3 1/2"                                                                                |
| Border 0",-4'-4"                   | Vintage Dark Hickory square | 4'-4 15/16" | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 0",0"                       | Vintage Dark Hickory square | 4'-4 15/16" | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 4'-4",0"                    | Vintage Dark Hickory square | 6'-1"       | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 4'-4",6'-0"                 | Vintage Dark Hickory square | 2'-4 15/16" | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 6'-8",6'-0"                 | Vintage Dark Hickory square | 5'-1"       | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 6'-8",11'-0"                | Vintage Dark Hickory square | 2'-8 15/16" | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 8'-11",11'-4 13/16"         | Vintage Dark Hickory square | 2'-8 13/16" | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 11'-2",11'-8 7/8"           | Vintage Dark Hickory square | 2'-8 11/16" | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 13'-5",12'-0 1/8"           | Vintage Dark Hickory square | 2'-8 5/8"   | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 15'-8",12'-2 3/4"           | Vintage Dark Hickory square | 2'-8 9/16"  | 1   | mitre at the corners; dividers on the doubled joist                                                                           |
| Border 17'-11",12'-4 9/16"         | Vintage Dark Hickory square | 2'-8 1/2"   | 1   | mitre at the corners; dividers on the doubled joist                                                                           |

| MEMBER                     | STOCK                       | LENGTH       | QTY | NOTE                                                                                  |
|----------------------------|-----------------------------|--------------|-----|---------------------------------------------------------------------------------------|
| Border 20'-2",12'-5 5/8"   | Vintage Dark Hickory square | 2'-4"        | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 22'-5",12'-5 5/8"   | Vintage Dark Hickory square | 2'-4"        | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 24'-8",12'-4 9/16"  | Vintage Dark Hickory square | 2'-8 1/2"    | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 26'-11",12'-2 3/4"  | Vintage Dark Hickory square | 2'-8 9/16"   | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 29'-2",12'-0 1/8"   | Vintage Dark Hickory square | 2'-8 5/8"    | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 31'-5",11'-8 7/8"   | Vintage Dark Hickory square | 2'-8 11/16"  | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 33'-8",11'-4 13/16" | Vintage Dark Hickory square | 2'-8 13/16"  | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 35'-11",11'-0"      | Vintage Dark Hickory square | 2'-8 15/16"  | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Border 38'-2",11'-0"       | Vintage Dark Hickory square | 13'-2 15/16" | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Breaker dw/m               | Vintage Dark Hickory square | 12'-7"       | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Breaker dm/s               | Vintage Dark Hickory square | 19'-11 5/8"  | 1   | mitre at the corners; dividers on the doubled joist                                   |
| Stair treads (wall:L)      | Vintage Coastline square    | 4'-0"        | 22  | 2 boards per tread, nosing 3/4" over the riser                                        |
| Stair risers (wall:L)      | riser board                 | 4'-0"        | 13  | rip to 7.38" less tread thickness                                                     |
| Left fascia (l)            | fascia                      | 4'-4 1/2"    | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (l)           | fascia                      | 4'-4 1/2"    | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (l)           | fascia                      | 4'-5"        | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (p)            | fascia                      | 10'-4 1/2"   | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (p)           | fascia                      | 10'-4 1/2"   | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (p)           | fascia                      | 2'-5"        | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (p2)           | fascia                      | 12'-0"       | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (p2)           | fascia                      | 4'-10 1/2"   | 1   | flush under the board nose; full board at the front corners, short piece at the house |

| MEMBER            | STOCK  | LENGTH     | QTY | NOTE                                                                                  |
|-------------------|--------|------------|-----|---------------------------------------------------------------------------------------|
| Right fascia (p2) | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (p2) | fascia | 4'-10 1/2" | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (p2) | fascia | 4'-4"      | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (w)   | fascia | 6'-6 1/2"  | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (w)  | fascia | 6'-6 1/2"  | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (w)  | fascia | 7'-5"      | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (m)   | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (m)   | fascia | 6 1/2"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (m)  | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (m)  | fascia | 6 1/2"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (m)  | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (m)  | fascia | 8'-0"      | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (s)   | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (s)   | fascia | 6'-6 1/2"  | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (s)  | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (s)  | fascia | 6'-6 1/2"  | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (s)  | fascia | 2'-2"      | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (a)   | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (a)   | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (a)   | fascia | 2'-6 1/2"  | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (a)  | fascia | 12'-0"     | 1   | flush under the board nose; full board at the front corners, short piece at the house |

| MEMBER           | STOCK  | LENGTH    | QTY | NOTE                                                                                  |
|------------------|--------|-----------|-----|---------------------------------------------------------------------------------------|
| Right fascia (a) | fascia | 12'-0"    | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (a) | fascia | 2'-6 1/2" | 1   | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (a) | fascia | 11'-2"    | 1   | flush under the board nose; full board at the front corners, short piece at the house |

THE STACK internal

|                                          |                                         |
|------------------------------------------|-----------------------------------------|
| Materials (raw)                          | \$39,873.29                             |
| Materials × 1.15                         | \$45,854.28                             |
| Sales tax 7.90% on factored materials    | \$3,622.49                              |
| Labor (rate card)                        | \$22,356.20                             |
| Work — carries the margin                | \$71,832.97                             |
| General conditions at cost               | \$2,966.00                              |
| Cost                                     | \$74,798.97                             |
| Sell — check or ACH at 43.0% GM          | \$128,893 · \$173.83/SF                 |
| Sell at the 40% floor                    | \$122,688                               |
| Financed                                 | \$138,595 · \$1,608/mo at 6.99% × 10 yr |
| Gross profit                             | \$54,094                                |
| Engineering — outside the price, at cost | \$1,800–\$3,000                         |

Labor

| LINE                                                  | QTY   | UNIT  | RATE     | EXT        |
|-------------------------------------------------------|-------|-------|----------|------------|
| Demolition — existing deck                            | 700   | SF    | \$2.00   | \$1,400.00 |
| Frame                                                 | 741.5 | SF    | \$10.00  | \$7,415.00 |
| Decking & fascia                                      | 741.5 | SF    | \$8.00   | \$5,932.00 |
| Railing                                               | 69.7  | LF    | \$20.00  | \$1,394.00 |
| Extra beam set                                        | 2     | set   | \$200.00 | \$400.00   |
| Stairs (wall:L) — 2 flights                           | 13    | riser | \$150.00 | \$1,950.00 |
| Landing guard rail (wall:L stair)                     | 21    | LF    | \$20.00  | \$420.00   |
| Stair landing 4'-0" x 4'-0" — frame, deck, 4 footings | 16    | SF    | \$58.00  | \$928.00   |
| Stair landing 7'-8" x 5'-8" — frame, deck, 4 footings | 43.4  | SF    | \$58.00  | \$2,517.20 |

General conditions — at cost, no margin

| ITEM               | AMOUNT  | WHY               |
|--------------------|---------|-------------------|
| General conditions | \$2,966 | \$4/SF x 741.5 SF |

What the price includes — sell values

| ITEM                    | VALUE     |
|-------------------------|-----------|
| Demolition & haul-off   | \$2,590   |
| Foundation & structure  | \$44,335  |
| Decking & fascia        | \$48,387  |
| Railing                 | \$9,403   |
| Stairs                  | \$21,047  |
| Site & project services | \$3,131   |
| Total, check or ACH     | \$128,893 |
| Total, financed         | \$138,595 |

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

GLENGARRY PL UPPER DECK — 1: FULTON RAIL · EXISTING COLUMNS AND CAISSONS

2x12 SYP frame at 12", 6x12 beam, 8x8 posts — on the 9 existing columns and caissons. TimberTech Vintage Coastline decking with a Dark Hickory border, Fulton rail.

**741.5 SF** · 7 zones · **12'-0"** above grade, engineered · **30 psf** snow design

Prepared for the owner of 166 Glengarry Pl, Castle Rock

PRICE

**Total investment \$138,595** — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about \$1,608/month

**Check or ACH: \$128,893** (7% savings)

WHAT THE PRICE INCLUDES

| ITEM                       | VALUE            |
|----------------------------|------------------|
| Demolition & haul-off      | \$2,590          |
| Foundation & structure     | \$44,335         |
| Decking & fascia           | \$48,387         |
| Railing                    | \$9,403          |
| Stairs                     | \$21,047         |
| Site & project services    | \$3,131          |
| <b>Total, check or ACH</b> | <b>\$128,893</b> |
| <b>Total, financed</b>     | <b>\$138,595</b> |

**Engineering — \$1,800 to \$3,000.** Stamped drawings by a Colorado engineer (30 psf snow). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

INCLUDED — WHAT IT IS

| ELEMENT | SELECTION                                                                                                                                                                                                                                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Decking | TimberTech PVC Vintage, Coastline; picture frame and dividers in Vintage Dark Hickory. Square-shoulder boards, 3/16" gaps, matching fascia. Ignition Resistant, Class A, WUI; Limited Lifetime product / 50-yr fade & stain.                                                                                                                                      |
| Railing | TimberTech Fulton Rail, Black, 36". 69.7 LF.                                                                                                                                                                                                                                                                                                                      |
| Frame   | 57 2x12 #1 True Frame Joist SYP GC @ 12" OC; 2-ply 2x12 front and sides and rear (freestanding); DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0 1/2" / 5'-7 1/2" (worst 1,394 lb); DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6" (worst 5,409 lb); DROP 6x12 DF A: 11'-0", 2 posts at 41'-3 1/2" / 50'-3 1/2" (worst 3,320 lb). |

## ELEMENT SELECTION

|             |                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Foundation  | The 9 existing stucco columns and caissons stay; new post bases epoxy-set on the caissons.                                                                   |
| Design load | 30 psf snow, stamped by a Colorado engineer. Colorado Wildfire Resiliency Code practice: Class A decking, noncombustible rail, metal flashing at every wall. |
| Hardware    | Simpson ZMAX galvanized connectors. Cortex fasteners with color-matched plugs.                                                                               |

## PLAN

7 zones · 741.5 SF · 51'-4" along the house · 12'-0" above grade · 69.7 LF Black rail

- **L · Stair pocket — 4'4" x 4'4" past the house's west wall; the stair leaves its north edge along the wall — 4'-4" x 4'-4".**
- **P · West pocket — 2'4" beside the bay, ledgered to the recessed wall — 2'-4" x 10'-4".**
- **P2 · West strip — 4'3" from the recessed wall to the curved front — 4'-3" x 16'-10".**
- **W · In front of the bay window — 7'4" x 5', the bay projects 6' (multi-angle facets in the model) — 7'-4" x 6'-6".**
- **M · Main run — 19'11" wall, 11' deep — 19'-11" x 12'-6".**
- **S · Step — 2'1" at the 7'6" jog — 2'-1" x 18'-6".**
- **A · Alcove bay — 11'1" into the 17' alcove, 26'6" deep, ends at the east wing wall — 11'-1" x 26'-6", no rail on the wall side.**

Zones are named left to right standing in the yard facing the house.

- Stair wall:L: 13 risers @ 7.38", 11 treads, 5 stringers, 4'-0" wide — 7 risers + 6 risers (left); landings 4'-0" x 4'-0" + 7'-8" x 5'-8"; landing guard 21 LF

## SCHEDULE

1. Engineering & ordering (2–3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811), old deck out.
3. Day 2 — layout, 9 Diamond Piers driven, ledger set.
4. Days 3–7 — posts and beams set.
5. Days 8–13 — frame, flashing at every wall.
6. Days 14–17 — decking, borders first.
7. Days 18–19 — Fulton rail.
8. Day 20 — stairs.
9. Day 21 — final clean; end of construction.

## NEXT STEP

Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer and supplies the design package; you pay GSX the engineer's cost at no markup

and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

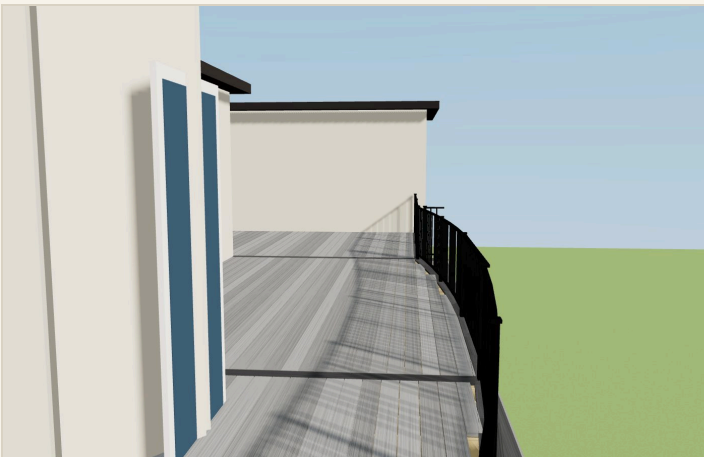
## RENDERS 3D model · design intent · timbers shown as specified



From the yard



From the front corner



Standing on the deck



Under the deck — posts, beams, footings

## ISOMETRICS



Isometric from the stair corner



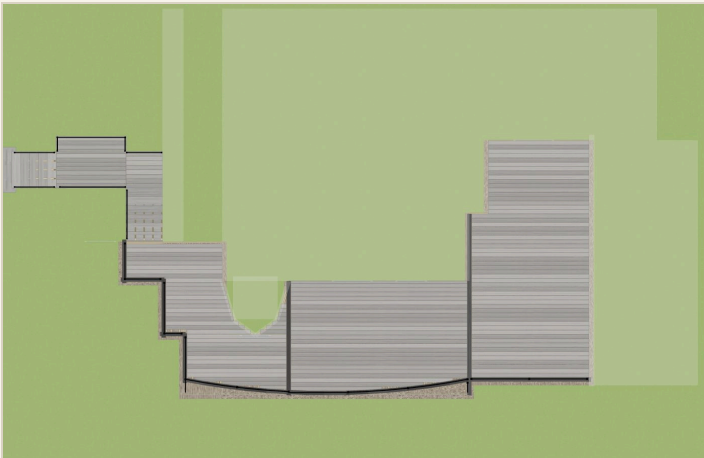
Isometric from the open corner



Low isometric

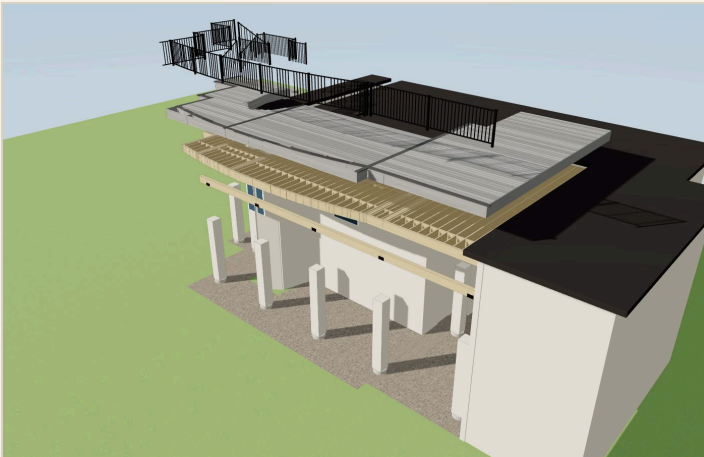


Isometric from under the frame



Plan from above

**EXPLODED**



Exploded — every layer



Exploded — from the open corner



Exploded — low angle

**DRAWING SET** G · S · A · D sheets — permit set with G-001 on top

## G-001 General notes · design criteria

## G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

Glengarry PI upper deck — 1: Fulton rail · existing columns and...

GSX · Jade Pender 303-550-9558 · September 21, 2026 · Rev 0 FOR REVIEW

## PROJECT

Glengarry Pl upper deck — 1: Fulton rail · existing columns and caissons  
166 Glengarry Pl, Castle Rock CO 80108  
741.5 SF · 12'-0" above grade · 7 zones

## DESIGN CRITERIA

|                          |                                                                                                                                                                                                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deck live load</b>    | 40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf                                                                                                                                                                                                             |
| <b>Ground snow</b>       | 30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs                                                                                                                                                                                                |
| <b>Dead load</b>         | 10 psf deck                                                                                                                                                                                                                                                                        |
| <b>Roof snow (cover)</b> | —                                                                                                                                                                                                                                                                                  |
| <b>Design total</b>      | 50 psf on the deck                                                                                                                                                                                                                                                                 |
| <b>Wind</b>              | 115 mph Vult, Exposure C, Risk Cat. II (ASCE 7-16) — confirm special wind region                                                                                                                                                                                                   |
| <b>Frost depth</b>       | 3'-0" (footings bear below)                                                                                                                                                                                                                                                        |
| <b>Soil bearing</b>      | 1,500 psf presumptive (IRC R401.4.1) — no soils report                                                                                                                                                                                                                             |
| <b>Seismic</b>           | B                                                                                                                                                                                                                                                                                  |
| <b>Wildfire (WUI)</b>    | Ignition-resistant construction: Class A / WUI-listed decking (TimberTech Advanced PVC), noncombustible rail (steel), fiber-cement skirt, soffit and fascia, 26 ga flashing at every wall, Class A roof on the cover; under-deck kept clear of combustibles (Colorado Springs WUI) |
| <b>Codes</b>             | 2021 IRC as adopted by Castle Rock; IRC R507 decks; ASCE 7-16 loads; IWUIC where mapped; Tables R507.5 / R507.6 / R507.7 / R507.9                                                                                                                                                  |
| <b>Engineering</b>       | Stamped structural set by a Colorado PE required — the stamped set governs                                                                                                                                                                                                         |

## STRUCTURE

|          |                                                                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Footings | the caissons                                                                                                                                                                                                                         |
| Posts    | the 9 existing stucco columns and caissons stay; new post bases epoxy-set                                                                                                                                                            |
| Beams    | DROP 6x12 DF L+P: 6'-7"; 2 posts at 1'-0 1/2" / 5'-7 1/2" (worst 1,394 lb); DROP 6x12 DF P2+WM+MS+A: 44'-7"; 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6" (worst 5,409 lb); DROP 57 2x12 #1 True Frame Joist SYP GC @ 12" OC |
| Joists   | 2x12 2x12 front and sides and rear (freestanding)                                                                                                                                                                                    |
| Rims     | 2x12 ledger x 47'-0" total - LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end - 48 fasteners                                                                                                            |
| Ledger   |                                                                                                                                                                                                                                      |
| Decking  | TimberTech Vintage Coastline — 55 rows parallel to the house, picture frame & dividers in Vintage Dark Hickory, 3/16" gaps, Cortex                                                                                                   |
| Rail     | Fulton 36" Black: 13 bays, 17 posts, 69.7 LF                                                                                                                                                                                         |

## SHEET INDEX

|       |                                 |
|-------|---------------------------------|
| G-001 | General notes - design criteria |
| A-001 | Isometric - renders             |
| S-100 | Foundation plan                 |
| S-101 | Framing plan                    |
| A-101 | Decking plan                    |
| A-201 | Railing plan + elevation        |
| A-301 | Section + front elevation       |
| D-501 | Details                         |
| T-400 | Build steps (build set)         |
| T-700 | Order + schedules (takeoff)     |

## SITE

Castle Rock (Douglas County) — combined sales tax 7.9%. Upper deck 12' above the lower patio; the stucco parapet, the stucco columns and the caissons are existing. Ponderosa lot — WUI, verify with the Town.

**FLAGS (INTERNAL — remove before the permit submittal)**

```
ENGINEER DRA 6'x12' D6: 6x12 not in IRC R507.5 – treated as 3/2x12; ...
ENGINEER cantilever 8'x14" of the 0" back splay – owner-directed, in ...
ENGINEER deck surface 12'-0" above grade – over 10'; lateral bracing and ...
ENGINEER existing caissons reused – the engineer verifies capacity for the ...
ENGINEER existing stout columns reused – the engineer verifies the column ...
CODE posts over 8' (9'-5" depth) – add diagonal knee bracing at each ...
CODE freestanding deck – no ledger lateral tie; provide diagonal ...
CODE floor zone: Vintage is Ignition Resistant, Class A, WUI – keep the ...
CODE wall, 1 stair: 13 risers – graspable handrail 34"–38" on at least ...
CODE wall, 1 stair landing: 36" min in the direction of travel and at ...
```

No scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

P14



P16



P10



P11



P15



P12



P13



OPEN — L (rear bldg) — P2

P1



P2



ROP 6x12 DF L+P

HOUSE — A

HOUSE — S

HOUSE — M

P8



P9



P3



P4



P5



P6



P7



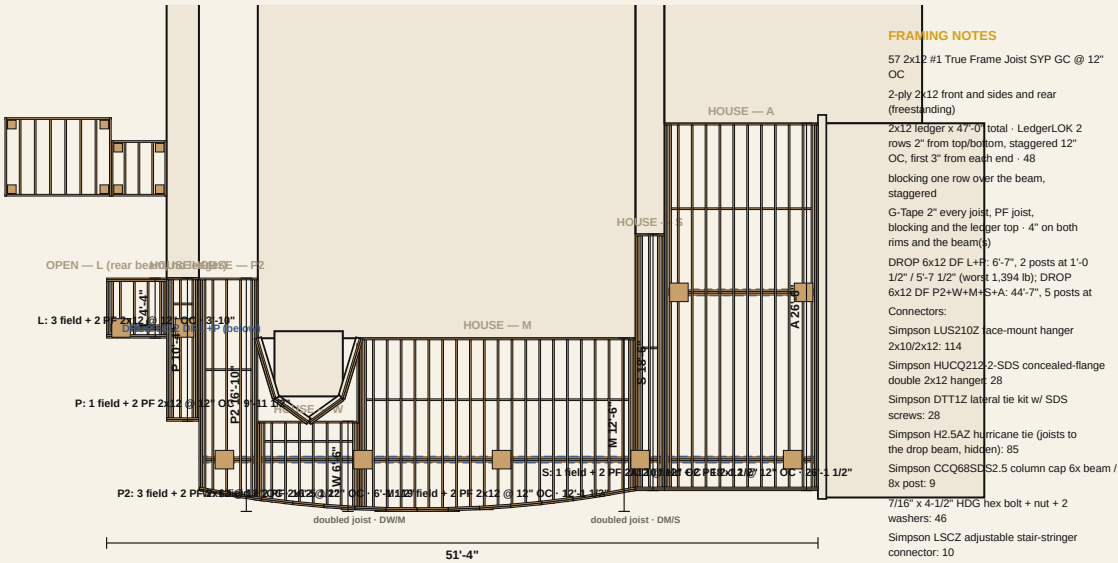
FOOTING SCHEDULE

Existing caissons and columns stay — verify plumb, bearing and condition; new column caps (engineered)

| #   | X          | Y             | ZONE  | LOAD     |
|-----|------------|---------------|-------|----------|
| P1  | 1'-0 1/2"  | -8 13/16"     | L     | 593 lb   |
| P2  | 5'-7 1/2"  | -8 13/16"     | P     | 1,394 lb |
| P3  | 8'-6"      | 8'-9 3/16"    | P2    | 3,470 lb |
| P4  | 18'-6"     | 8'-9 3/16"    | M     | 4,026 lb |
| P5  | 28'-6"     | 8'-9 3/16"    | M     | 4,026 lb |
| P6  | 38'-6"     | 8'-9 3/16"    | S     | 5,409 lb |
| P7  | 49'-6"     | 8'-9 3/16"    | A     | 3,014 lb |
| P8  | 41'-3 1/2" | -3'-3 5/8"    | A     | 3,320 lb |
| P9  | 50'-3 1/2" | -3'-3 5/8"    | A     | 3,320 lb |
| P10 | 10"        | -13'-8 15/16" | stair | landing  |
| P11 | 10"        | -10'-8 15/16" | stair | landing  |
| P12 | 3'-10"     | -13'-8 15/16" | stair | landing  |
| P13 | 3'-10"     | -10'-8 15/16" | stair | landing  |
| P14 | -6'-10"    | -15'-4 15/16" | stair | landing  |
| P15 | -6'-10"    | -10'-8 15/16" | stair | landing  |
| P16 | -2"        | -15'-4 15/16" | stair | landing  |
| P17 | -2"        | -10'-8 15/16" | stair | landing  |

x from the left frame face facing the house; y from the reference wall.  
Frost 3'-0" - soil 1,500 psf - worst post 5,409 lb

51'-4" along the house  
L 4'-4" P 2'-4" P2 4'-3" W 7'-4" M 19'-11" S 2'-1" A 11'-1"



A-101 Decking plan

A-101

DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

Glengarry PI upper deck — 1: Fulton rail · existing columns and...

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Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

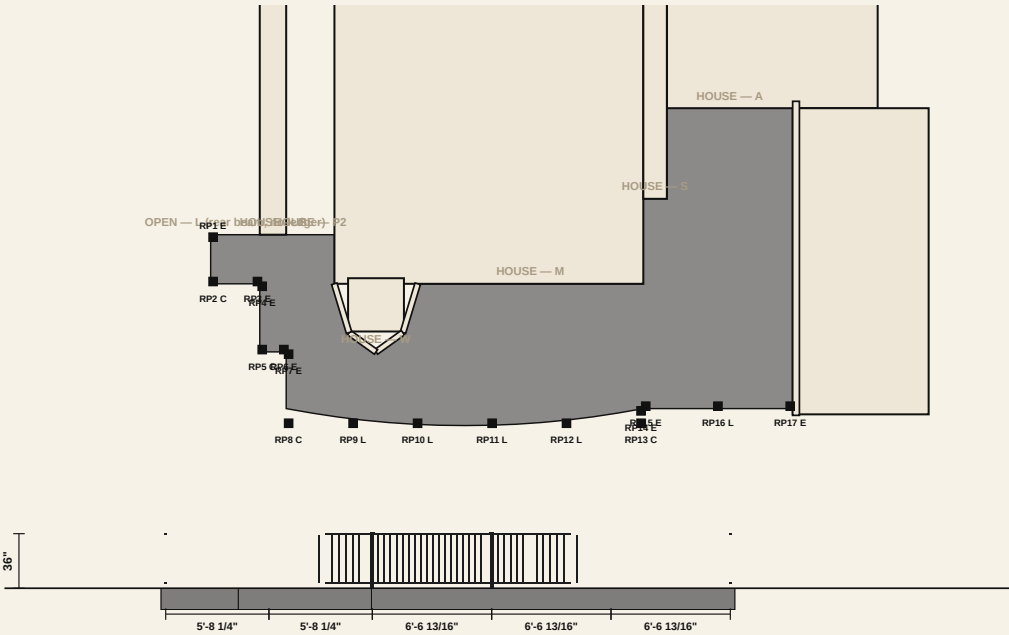
RAILING

Fulton 36" Black: 13 bays, 17 posts,  
69.7 LF

Fulton 36" — 17 posts (4 CORNER, 8 END,  
5 LINE) inside the outer rim ply, inner  
ply pocketed 2" wide, (2) 7/16" x 4-1/2"  
bolts per post

Bays: end:left 3'-11" · front:L 3'-11" ·  
step:L-P 5'-7" · front:P 1'-11"  
step:P-P2 6'-1" · front:P2+W+M 5'-8 1/4"  
· front:P2+W+M 5'-8 1/4" · front:P2+W+M  
6'-6 13/16" · front:P2+W+M 6'-6 13/16" ·  
front:P2+W+M 6'-6 13/16" · step:M-S

Guard 36" min above the deck; openings  
under 4" (IRC R312). Rail LF for labor =  
deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

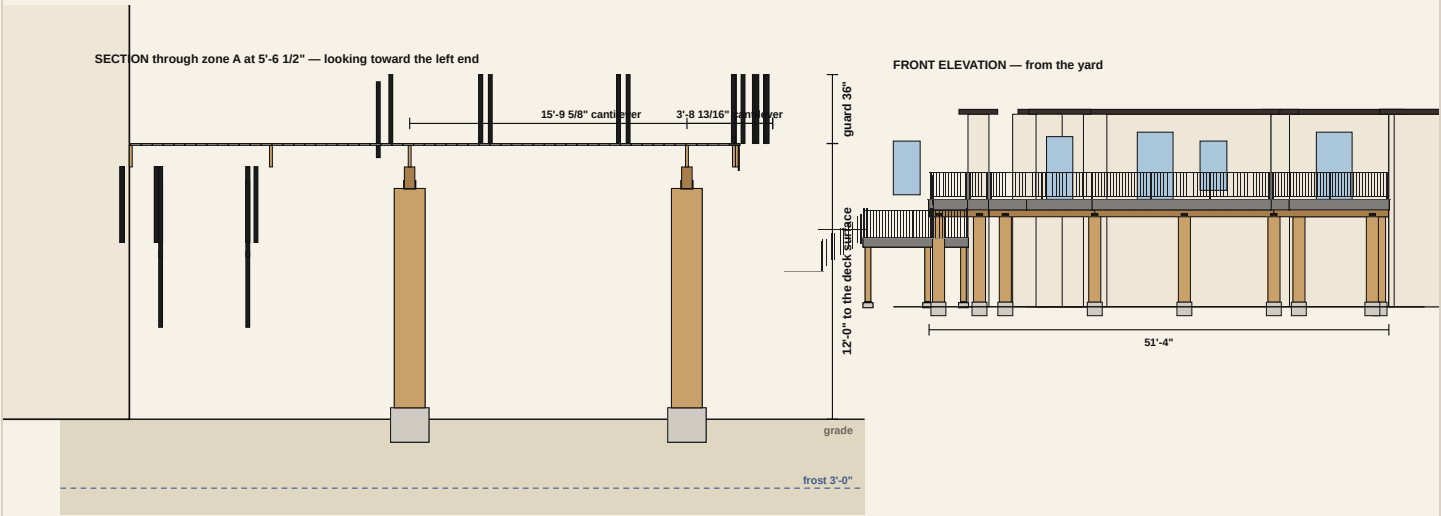
A-301 Section + front elevation

A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

Glengarry PI upper deck — 1: Fulton rail · existing columns and...

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SECTION NOTES

Deck surface 12'-0" above grade; guard 36" where the drop exceeds 30"

DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0 1/2" / 5'-7 1/2" (worst 1,394 lb); DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6" (worst 5,409 lb);

Posts the 9 existing stucco columns and caissons stay; new post bases epoxy-set on the caissons

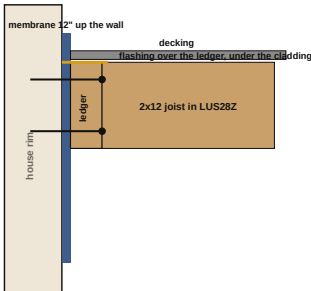
2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

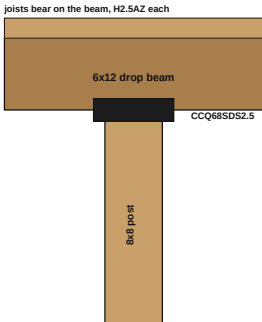
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

1 LEDGER AT THE HOUSE



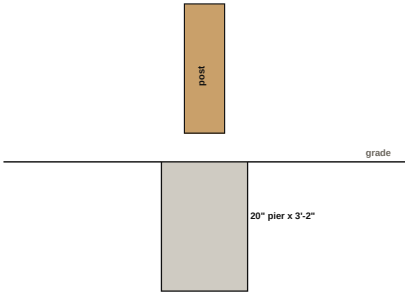
2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners · 26 ga wall flashing (WUJ)

2 POST · BEAM · CAP

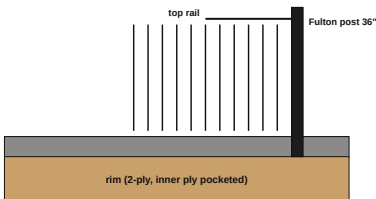


DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0" / 2' / 5'-7" / 1/2" (worst 1,394 lb); DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-

3 FOOTING · POST BASE

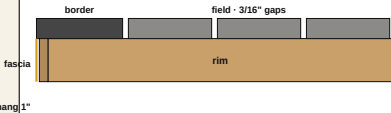


4 RAIL POST · GUARD



Fulton 36" — 17 posts (4 CORNER, 8 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

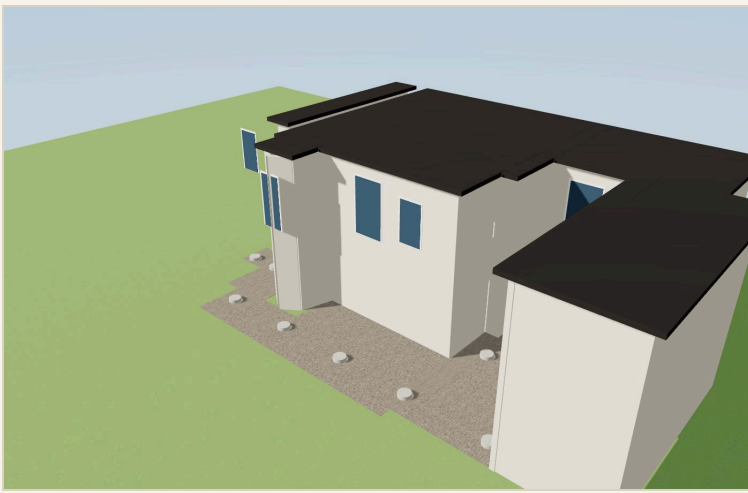
5 DECK EDGE · BORDER



Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (3454 field + 232 border/divider)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.



Step 1 · Footings — 9 existing caissons stay with the existing columns

## T-400 · Step 1 · Footings

### GOES IN

### HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: DROP 6x12 DF L+P at -8 13/16" from the reference wall, posts at 1'-0 1/2" / 5'-7 1/2"; DROP 6x12 DF P2+W+M+S+A at 8'-9 3/16" from the reference wall, posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6"; DROP 6x12 DF A at -3'-3 5/8" from the reference wall, posts at 41'-3 1/2" / 50'-3 1/2".
- The 9 existing caissons and columns stay. Expose each caisson top, check plumb and cracking, photograph; the engineer confirms the column and caisson before the new beams land on them.

### CHECK

- Every existing caisson sound, level and on the beam line; engineer's sign-off in the file
- Epoxy cured before load
- Photos before backfill



Step 2 · Posts — 8x8 #2 GC on ZMAX bases

## T-401 · Step 2 · Posts

### GOES IN

### HOW

- The existing stucco columns stay. Strip the old beam off each column top, check the column core and its bearing; new column caps / saddles per the engineer, then the beams land on them.

### CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff

## T-403 · Step 4 · Beams

### GOES IN

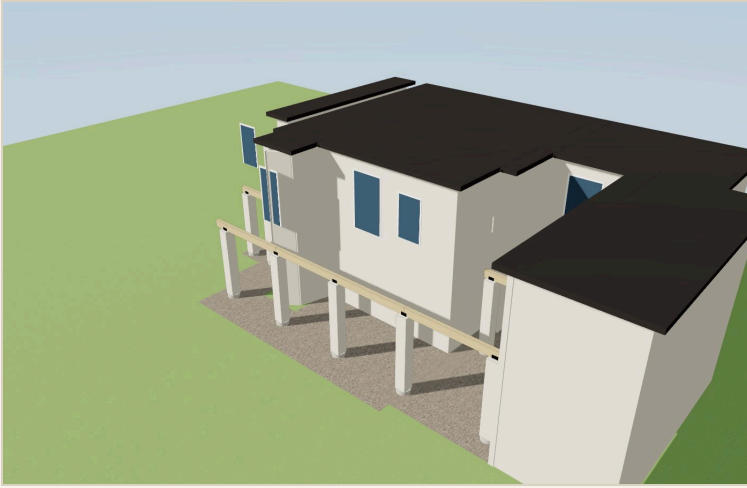
- (2) 2x6x12 #1 True Frame Joist SYP GC
- (3) 6x12x12 Douglas Fir #2
- (1) 6x12x20 Douglas Fir #2
- (1) 6x12x16 Douglas Fir #2
- (9) Simpson CCQ68SDS2.5 column cap 6x beam / 8x post

### HOW

- DROP 6x12 DF L+P: 6'-7" one piece, 2 posts @ 4'-7" max
- DROP 6x12 DF P2+W+M+S+A: 44'-7" in 3 pieces spliced over posts, 5 posts @ 11'-0" max
- DROP 6x12 DF A: 11'-0" one piece, 2 posts @ 9'-0" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.

### CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened



Step 4 · Beams — DROP 6x12 DF L+P (2 posts) · DROP 6x12 DF P2+W+M+S+A (5 posts) · DROP 6x12 DF A (2 posts)

## T-404 · Step 5 · Frame

### GOES IN

- (12) 2x12x10 #1 True Frame Joist SYP GC
- (30) 2x12x12 #1 True Frame Joist SYP GC
- (26) 2x12x14 #1 True Frame Joist SYP GC
- (20) 2x12x20 #1 True Frame Joist SYP GC
- (16) 2x12x24 #1 True Frame Joist SYP GC
- (3) 6x12x12 Douglas Fir #2
- (114) Simpson LUS210Z face-mount hanger 2x10/2x12
- (28) Simpson HUCQ212-2-SDS concealed-flange double 2x12 hanger
- (3) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (3) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (85) Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)
- (2) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (8) Aluminum L-flashing 10' (over ledger + side rims)

### HOW

- 2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners
- L: 3 joists @ 12" OC x 3'-10"; P: 1 joists @ 12" OC x 9'-11 1/2"; P2: 3 joists @ 12" OC x 16'-5 1/2"; W: 6 joists @ 12" OC x 6'-1 1/2"; M: 19 joists @ 12" OC x 12'-1 1/2"; S: 1 joists @ 12" OC x 18'-1 1/2"; A: 10 joists @ 12" OC x 26'-1 1/2". Crown up, hangers at the ledger.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

### CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 5 · Frame — 2x12 joists at 12", rims, ledger, blocking over the beam

## T-405 · Step 6 · Decking

### GOES IN

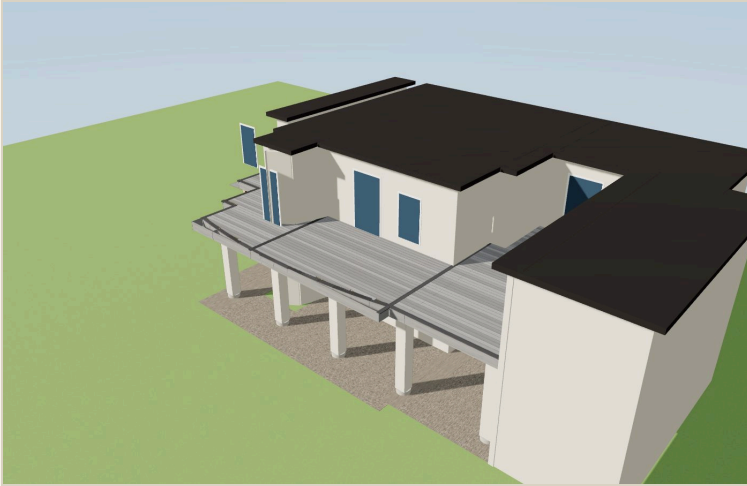
- (30) TimberTech Vintage Coastline 1x6x12 Square Edge
- (30) TimberTech Vintage Coastline 1x6x16 Square Edge
- (40) TimberTech Vintage Coastline 1x6x20 Square Edge
- (1) TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders & dividers
- (5) TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders & dividers
- (21) TimberTech Vintage Fascia Coastline 0.5x11.75x12
- (10) Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)

### HOW

- Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (3454 field + 232 border/divider)
- Borders first, then the field.
- Rows per panel: panel 1 (0"-18'-3") 34 rows · panel 2 (18'-3"-38'-2") 26 rows · panel 3 (38'-2"-51'-4") 55 rows · 3 ripped pieces · 2 breakers. Borders and breakers first, then the field rows from the outer edge toward the house.

### CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 6 · Decking — TimberTech Vintage Coastline, Dark Hickory border, hidden fasteners

## T-406 · Step 7 · Rail

### GOES IN

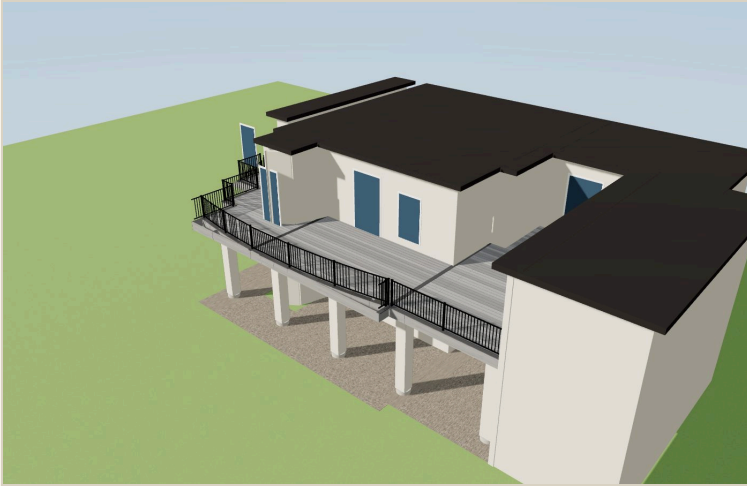
- (7) TimberTech Fulton Rail 6' x 36" level panel Black
- (6) TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)
- (4) TimberTech Fulton 2" CORNER post 36" Black w/ brackets, cap, skirt
- (8) TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt
- (5) TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt
- (1) Rust-Oleum flat black touch-up (cut ends)

### HOW

- Fulton 36" — 17 posts (4 CORNER, 8 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

### CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction



Step 7 · Rail — Fulton 36" Black



The interactive 3D model lives at the online version of this document.

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GS Exterior Experts · Deck Division

Prices: rate card · GC at cost · 42.5% GM · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.